



Paper 78 summary

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: singhsamikshasingh2004@gmail.com

Sun, 26 Apr 2026 at 4:23pm

Hello.

Here is submission summary.

Track Name: Agricultural, Environmental, and Life Sciences

Paper ID: 78

Paper Title: Performance Evaluation of CNN and MobileNetV2 for Mango Leaf Disease

Abstract:
When mango leaves get sick, it doesn't just hurt the harvest, it can also ruin the whole crop. Thus, before the diseases spread and do real harm the farmers need to find them early. It always took too long to check each leaf by hand additionally, many mistakes happened. So we were sure that there has to be a better way. We tried two different ways through which we could automatically find diseases on mango leaves. Firstly, we implemented our own neural network (CNN) from the ground up. Second, we used MobileNetV2, which learns from models that are already out there instead of starting from scratch. We searched and gathered 4,000 leaf images of eight different diseases to check whether these methods worked or not. Then we separated them for training, testing, and validation. We also used some tricks so that we can help the models to deal with new different pictures they hadn't seen before. The results were pretty clear: MobileNetV2 got 99.4% of the answers right, while our own CNN got about 97% of them right. When we looked more closely at the results using precision, recall, and F1-score, both the models did pretty well, but MobileNetV2 was still the best. This really shows that transfer learning makes a big difference. MobileNetV2 is light, fast and works really well on mobile devices with limited sources on it and this is really great for real - world farming. It is very quick and accurate and finds diseases even when the field conditions change. These all the advantages makes it very useful for farmers who need quick results of the diseases to protect their hardwork while they go to harvest the crops.

Created on: Sun, 26 Apr 2026 10:52:59 GMT

Last Modified: Sun, 26 Apr 2026 10:52:59 GMT

Authors:
- singhsamikshasingh2004@gmail.com (Primary)
- kavya.arora.ar@gmail.com
- pranyakhattar@gmail.com
- ankitakhatri699@gmail.com

Primary Subject Area: Botany علم النبات - plant pathology امراض نبات

Secondary Subject Areas:
Agricultural Sciences العلوم الزراعية - وقاية النبات والامن البيولوجي Plant Protection and Biosecurity

Submission Files:
Performance Evaluation of CNN and MobileNetV2 for Mango Leaf Disease.docx (910 Kb, Sun, 26 Apr 2026 10:49:38 GMT)

Submission Questions Response:

1. Full Name
Samiksha Singh
2. Second Author
Kavya Arora
3. 3th Author
Pranya Khattar
4. 4th Author
Ankita Khatri
5. 5th Author
[Not Answered]
6. Country
India
7. Organization
Chitkara University Punjab
8. Academic Rank
اخرى (Others)
9. Phone Number
9041709147
10. Comment
We are currently college students, and this is our first time applying for research.

Thanks,
CMT Team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)